Exploratory Data Analysis: Store Type A

2026-04-08

EDA Goals

We use this EDA to answer four modeling questions:

1. What does sales look like over time (stable, trend, or volatile)?
2. Is there repeating seasonality (weekly/monthly)?
3. Which factors affect sales (Promo, CompetitionDistance, DayOfWeek)?
4. Is the series stationary (to guide ARIMA differencing)?

```r
library(tidyverse)
library(lubridate)
library(tseries)
library(forecast)
```

Load Data

```r
agg <- read_csv("../data/processed/store_type_a_train_aggregate_daily.csv") %>%
  mutate(Date = as.Date(Date)) %>%
  arrange(Date)

glimpse(agg)
```

```
## Rows: 942
## Columns: 11
## $ Date               <date> 2013-01-01, 2013-01-02, 2013-01-03, 2013-01-04, 20~
## $ Sales              <dbl> 2907, 3920386, 3528027, 3610215, 2964587, 6254, 609~
## $ OpenStores         <dbl> 1, 602, 601, 601, 600, 3, 599, 599, 599, 599, 599, ~
## $ PromoStores        <dbl> 0, 0, 0, 0, 0, 0, 602, 602, 602, 602, 602, 0, 0, 0,~
## $ PromoShare         <dbl> 0.000000, 0.000000, 0.000000, 0.000000, 0.000000, 0~
## $ SchoolHolidayShare <dbl> 1.00000000, 1.00000000, 0.93853821, 0.93853821, 0.0~
## $ DayOfWeek          <dbl> 2, 3, 4, 5, 6, 7, 1, 2, 3, 4, 5, 6, 7, 1, 2, 3, 4, ~
## $ Year               <dbl> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 201~
## $ Month              <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ IsoWeek            <dbl> 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, ~
## $ IsWeekend          <dbl> 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, ~
```

Aggregate-Daily Summary

```r
summary(agg$Sales)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##       0 2994007 3563650 3360228 4459142 8511488
```

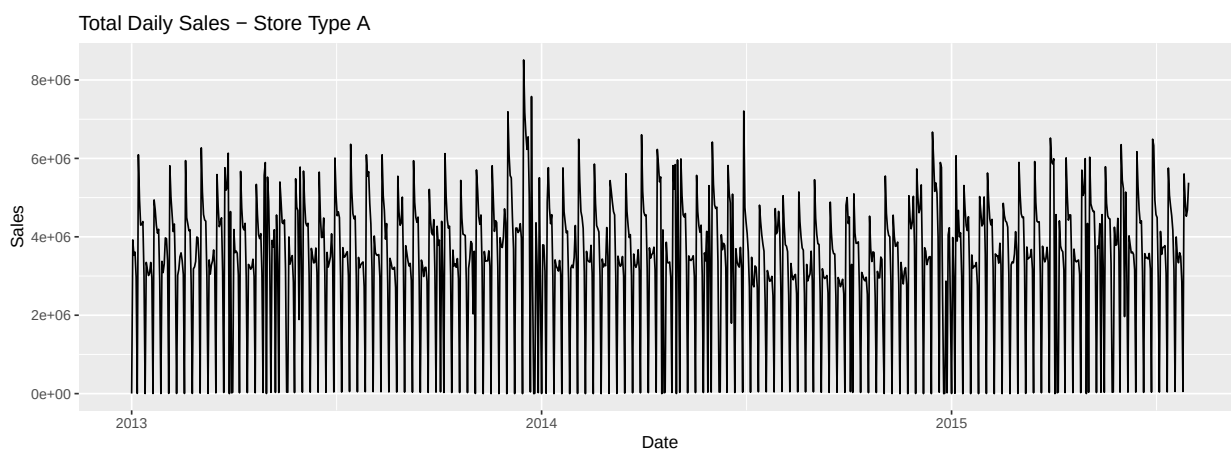
Date Continuity Check

```
date_span <- as.integer(max(agg$Date) - min(agg$Date)) + 1
missing_days <- date_span - nrow(agg)

tibble(
  first_date = min(agg$Date),
  last_date = max(agg$Date),
  expected_days = date_span,
  observed_days = nrow(agg),
  missing_days = missing_days
)
```

```
## # A tibble: 1 x 5
##   first_date last_date  expected_days observed_days missing_days
##   <date>     <date>             <dbl>         <int>        <dbl>
## 1 2013-01-01 2015-07-31           942           942            0
```

What Does Sales Look Like Over Time?

```
ggplot(agg, aes(x = Date, y = Sales)) +
  geom_line() +
  labs(
    title = "Total Daily Sales - Store Type A",
    x = "Date",
    y = "Sales"
  )
```



Promo Effect: Promo vs Non-Promo Days

```
promo_summary <- agg %>%
  mutate(PromoDay = if_else(PromoShare > 0, 1L, 0L)) %>%
  group_by(DayOfWeek, PromoDay) %>%
  summarise(
    mean_sales = mean(Sales, na.rm = TRUE),
    sd_sales = sd(Sales, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  )
```

```r
promo_summary
```

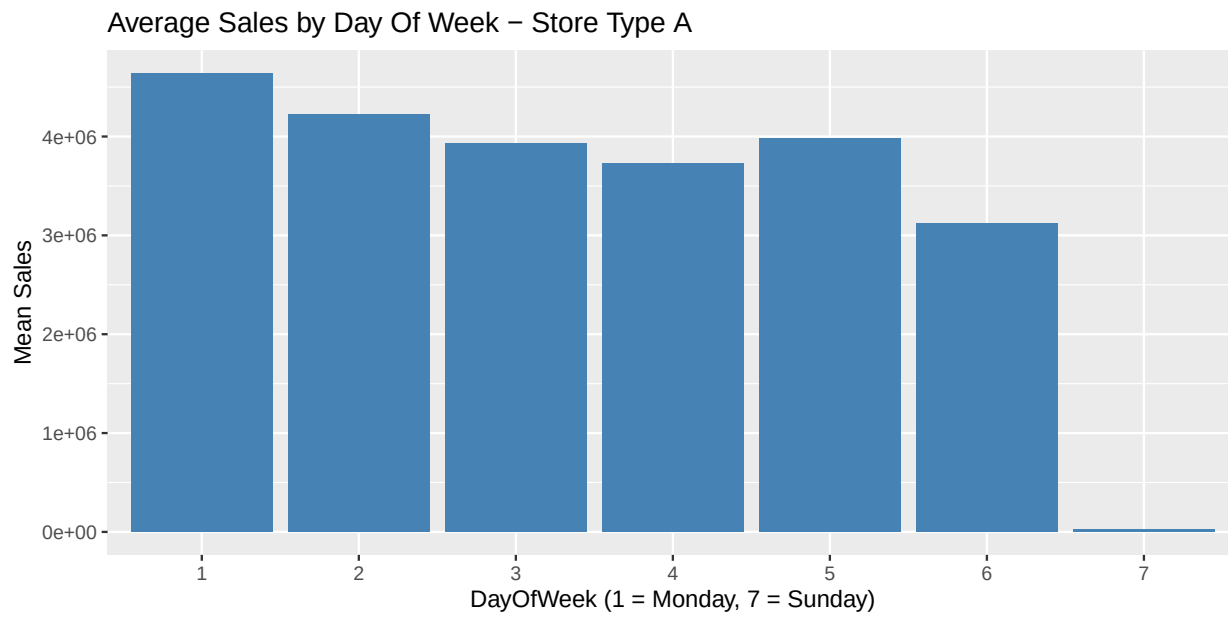
```
## # A tibble: 12 x 5
##    DayOfWeek PromoDay mean_sales sd_sales     n
##        <dbl>    <int>      <dbl>    <dbl> <int>
##  1         1        0   3352902. 1294884.    62
##  2         1        1   5751046.  654095.    72
##  3         2        0   3394326.  653628.    63
##  4         2        1   4943941.  575095.    72
##  5         3        0   3244104.  712786.    63
##  6         3        1   4473674.  755051.    72
##  7         4        0   3042221. 1110822.    63
##  8         4        1   4272478.  917407.    72
##  9         5        0   3442823. 1009332.    66
## 10         5        1   4327555.  921347.    69
## 11         6        0   3125231.  443886.   134
## 12         7        0     23905.   16795.   134
```

```r
promo_lm <- agg %>%
  filter(Sales > 0) %>%
  mutate(
    DayOfWeek = factor(DayOfWeek),
    Month = factor(Month)
  ) %>%
  lm(log1p(Sales) ~ PromoShare + OpenStores + DayOfWeek + Month, data = .)

coef(summary(promo_lm))["PromoShare", ]
```

```
##      Estimate    Std. Error       t value      Pr(>|t|)
## -4.667888e-03  4.520661e-04 -1.032568e+01  1.079745e-23
```

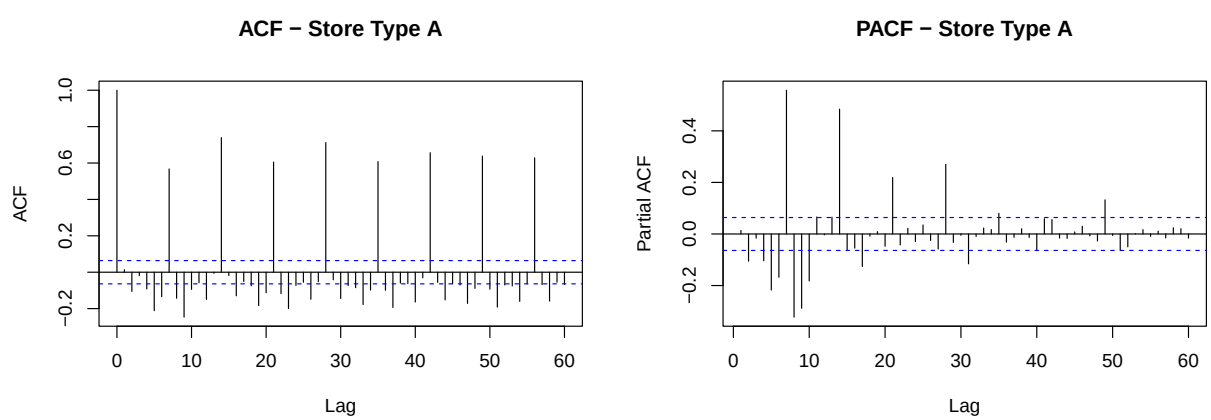
Day-of-Week Effect

```r
dow_summary <- agg %>%
  filter(Sales > 0) %>%
  group_by(DayOfWeek) %>%
  summarise(mean_sales = mean(Sales, na.rm = TRUE), .groups = "drop")

ggplot(dow_summary, aes(x = factor(DayOfWeek), y = mean_sales)) +
  geom_col(fill = "steelblue") +
  labs(
    title = "Average Sales by Day Of Week - Store Type A",
    x = "DayOfWeek (1 = Monday, 7 = Sunday)",
    y = "Mean Sales"
  )
```



ACF and PACF

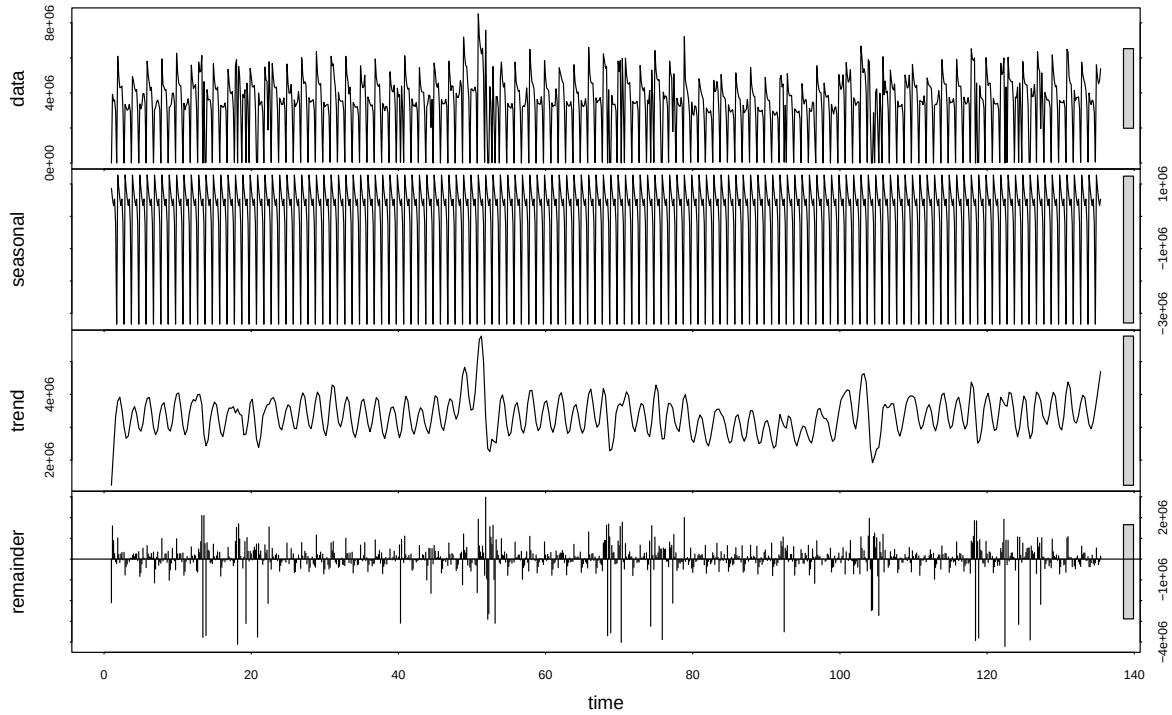
```
par(mfrow = c(1, 2))
acf(agg$Sales, lag.max = 60, main = "ACF - Store Type A")
pacf(agg$Sales, lag.max = 60, main = "PACF - Store Type A")
```




```
par(mfrow = c(1, 1))
```

STL Decomposition (Weekly Frequency)

```
sales_ts <- ts(agg$Sales, frequency = 7)
stl_fit <- stl(sales_ts, s.window = "periodic")
plot(stl_fit)
```



Stationarity Tests

```
adf_result <- adf.test(agg$Sales)
kpss_result <- kpss.test(agg$Sales)

adf_result
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  agg$Sales
## Dickey-Fuller = -15.968, Lag order = 9, p-value = 0.01
## alternative hypothesis: stationary
```

```
kpss_result
```

```
##
##  KPSS Test for Level Stationarity
##
## data:  agg$Sales
## KPSS Level = 0.12577, Truncation lag parameter = 7, p-value = 0.1
```

EDA Notes for Modeling

- Strong weekly behavior (period 7) supports seasonal ARIMA terms.
- Promo share, open stores, and day-of-week effects suggest useful external regressors for ARMAX.
- Stationarity test outcomes guide whether differencing is required.